

Elementary Problems

Double Counting

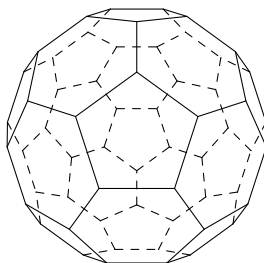
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Basic Work

1. Some students in the school join some clubs. Each student joins exactly 3 clubs, and each club consists of exactly 3 students. Prove that the number of students is the same as the number of clubs.
2. Let $A_1, A_2, \dots, A_m$ be subsets of $\{1, 2, \dots, n\}$ such that each subset has exactly 3 elements and $|A_i \cap A_j| \leq 1$ for any $i \neq j$. Prove that $m \leq \frac{n(n-1)}{6}$.
3. There are 8 students taking a test with 8 problems. Each problem is solved by exactly 5 students. Prove that there are 2 students such that every problem is solved by at least one of them.
4. (Iran MO 2nd Round 2010 Problem 6) A school has n students and some super classes are provided for them. Each student can participate in any number of classes that he/she wants. Every class has at least two students participating in it. We know that if two different classes have at least two common students, then the number of the students in the first of these two classes is different from the number of the students in the second one. Prove that the number of classes is not greater than $(n-1)^2$.
5. There are $2n$ students taking a test. It turns out that no matter how the students are partitioned into two groups of n people, for every question in the test, there is at least one student from each group who has answered that question correctly. Prove that there is at least one student who gets correct answers in more than half of the questions.
6. (All-Russian MO 1966) A group of children are arranged into two rows with the same number of children, so that every child in the back row is taller than the child standing in front of him in the first row. Prove that this remains true if each row is rearranged so that every child is taller than the one (if exists) on his left.
(Remark: We may assume the heights of the children are all distinct.)
7. There are 21 different dominoes. Each domino consists of two squares, each of which is marked with 1, 2, 3, 4, 5 or 6 dots. Prove that we cannot align all the dominoes in a circle so that the adjacent halves of neighbouring dominoes have the same number of dots.
8. There is an $m \times n$ chessboard whose cells are coloured black and white alternately. Each cell contains an integer. If the sum of numbers in each row and each column is even, prove that the sum of all numbers in the black cells is even.
9. (All-Russian MO 1965) A group of 100 people patrol the local streets so that in every evening there are 3 people on duty. Prove that it is not possible to make every pair of people to be on duty simultaneously for exactly one time.

10. (IMO shortlist 1975) There are six ports on a lake. Is it possible to organize a series of routes satisfying the following conditions?
- (i) Every route includes exactly three ports;
 - (ii) No two routes contain the same three ports;
 - (iii) The series offers exactly two routes to each tourist who desires to visit two different arbitrary ports.
11. Is it possible to write down the integers from 1 to 32 on the 32 faces of a football, so that the number written on each black face (pentagonal face) is equal to the sum of numbers written on its adjacent faces?



12. Let m and n be positive integers. Compute

$$\sum_{(A_1, A_2, \dots, A_m)} |A_1 \cup A_2 \cup \dots \cup A_m|$$

where the sum runs over all m -tuples of subsets of $\{1, 2, \dots, n\}$.

13. (IMO 1987 Problem 1) Let $p_n(k)$ be the number of permutations of the set $\{1, \dots, n\}$, $n \geq 1$, which have exactly k fixed points. Prove that

$$\sum_{k=0}^n k \cdot p_n(k) = n!.$$

(Remark: A permutation f of a set S is a one-to-one mapping of S onto itself. An element i in S is called a fixed point of the permutation f if $f(i) = i$.)

14. Let $n > 1$ be an integer. The cells of an $n \times n$ board are coloured black and white arbitrarily. Prove that we can remove one row and one column of the board such that the number of black cells and the number of white cells in the resultant board are different.

15. (Iran MO 2nd Round 2003 Problem 3) n volleyball teams have competed to each other (each 2 teams have competed exactly 1 time). For every 2 distinct teams like A, B , there exist exactly t teams which have lost their match with A, B . Prove that $n = 4t + 3$.
(Remark: There is no tie in a match. The t is fixed for any pair of teams.)
16. (USATST 2010 Problem 6) Let T be a finite set of positive integers greater than 1. A subset S of T is called good if for every $t \in T$ there exists some $s \in S$ with $\gcd(s, t) > 1$. Prove that the number of good subsets of T is odd.
17. (EGMO 2013 Problem 6) Snow White and the Seven Dwarves are living in their house in the forest. On each of 16 consecutive days, some of the dwarves worked in the diamond mine while the remaining dwarves collected berries in the forest. No dwarf performed both types of work on the same day. On any two different (not necessarily consecutive) days, at least three dwarves each performed both types of work. Further, on the first day, all seven dwarves worked in the diamond mine.
Prove that, on one of these 16 days, all seven dwarves were collecting berries.
18. (Brazil MO 2002 Problem 3) The squares of an $m \times n$ board are labelled from 1 to mn so that the squares labelled i and $i + 1$ always have a side in common. Show that for some k the squares k and $k + 3$ have a side in common.
(Remark: We may assume $m, n \geq 2$ since otherwise the assertion is not true.)
19. Let $m, n \geq 2$ be integers. Let $S_1, S_2, \dots, S_m$ be distinct subsets of $\{1, 2, \dots, n\}$ such that $|S_i \cap S_j| = 1$ for all $i \neq j$. Prove that $m \leq n$.

Binomial Coefficient

20. Let n be a positive integer. Prove that

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

21. Let m and n be positive integers such that $m \leq n$. Prove that

$$\sum_{k=m}^n \binom{k}{m} = \binom{n+1}{m+1}.$$

22. Let k, m, n be nonnegative integers such that $k \leq m, n$. Prove that

$$\binom{m+n}{k} = \sum_{j=0}^k \binom{m}{j} \binom{n}{k-j}.$$

23. Let m and n be nonnegative integers such that $m \leq n$. Prove that

$$\sum_{j=0}^n \binom{n}{j} \binom{n-j}{m-j} = 2^m \binom{n}{m}.$$

24. Let n be a nonnegative integer. Prove that

$$\sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} 2^{n-2j} \binom{n}{2j} \binom{2j}{j} = \binom{2n}{n}.$$

25. Let n be a nonnegative integer. Prove that

$$\sum_{j=1}^{n+1} \binom{n+1}{j} \sum_{k=0}^{j-1} \binom{n}{k} = 2^{2n}.$$

26. (IMO 1998 Problem 2) In a competition, there are a contestants and b judges, where $b \geq 3$ is an odd integer. Each judge rates each contestant as either “pass” or “fail”. Suppose k is a number such that, for any two judges, their ratings coincide for at most k contestants. Prove that $\frac{k}{a} \geq \frac{b-1}{2b}$.

27. Let n and k be positive integers. Let $A_1, A_2, \dots, A_{n+1}$ be subsets of a set A such that each A_i has n elements, the intersection of any pair A_i and A_j has at most k elements, and $A = \bigcup_{i=1}^{n+1} A_i$. Prove that $|A| \geq \frac{n(n+1)}{k+1}$.

28. (CWMO 2003 Problem 8) 1650 students are arranged in 22 rows and 75 columns. It is known that for any two columns, the number of pairs of students in the same row that have the same gender does not exceed 11. Prove that the number of boys does not exceed 928.

29. (Russian MO 1996 Grade 9 Problem 4) In the Duma there are 1600 delegates, who have formed 16000 committees of 80 persons each. Prove that one can find two committees having no fewer than four common members.

30. (USAMO 1976 Problem 1)

- (a) Suppose that each square of a 4×7 chessboard is coloured either black or white. Prove that with any such colouring, the board must contain a rectangle (formed by the horizontal and vertical lines of the board) whose four distinct unit corner squares are all of the same colour.
- (b) Exhibit a black-white colouring of a 4×6 board in which the four corner squares of every rectangle, as described above, are not all of the same colour.

31. (Peru 2006) All the squares of a board of $(n + 1) \times (n - 1)$ squares are painted with three colours such that, for any two different columns and any two different rows, the 4 squares in their intersections don't have all the same colour. Find the greatest possible value of n .
32. We place $n(n + 1) + 1$ rooks on a $2n \times 2n$ chessboard. Prove that there is a rook that is threatened by at least n other rooks. The rooks threaten each other if they are put on two cells in the same row or the same column. There could be some other rooks in between.

Record

v1.0: problems 1–32